

Macroeconomics

IS-LM-PC Model

(Blanchard, Macroeconomics 8th edition Ch. 9)

IS-LM-PC Model

- ▶ From the Short to the Medium Run:
 - ▶ First, we looked at equilibrium in the goods and financial markets, and how output is determined in the short run.
 - ▶ Then, we looked at equilibrium in the labor market.
 - ▶ We now put the two parts together and use this **IS-LM-PC** (PC for Phillips curve) model to characterize the behavior of output both in the short run and the medium runs.

Model Components

- ▶ Recall, output in the short run is determined by demand (IS curve):

$$Y = C(Y - T) + I(Y, r + x) + G$$

- ▶ Also, the Phillips curve:

$$\pi - \pi^e = -\alpha(u - u_n)$$

- ▶ Employment is

$$N = L(1 - u) \Rightarrow Y = N = L(1 - u)$$

- ▶ When $u = u_n$

- ▶ **Natural employment** is $N_n = L(1 - u_n)$
- ▶ **Potential output** is $Y_n = L(1 - u_n)$

Output Gap

- ▶ **Output gap:** the deviation of output from its natural level

$$Y - Y_n = L((1 - u) - (1 - u_n)) = -L(u - u_n) \\ \Rightarrow \pi - \pi^e = (\alpha/L)(Y - Y_n)$$

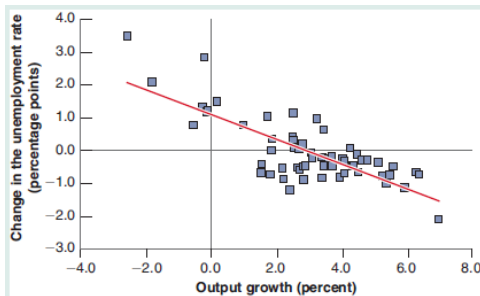
- ▶ Assuming that inflation expectations are anchored at the target rate of the central bank $\bar{\pi}$ gives:

$$\pi - \bar{\pi} = (\alpha/L)(Y - Y_n)$$

- ▶ Positive relation between output and inflation: When output is above potential (positive output gap), inflation exceeds target inflation, and vice versa.
- ▶ <https://www.youtube.com/watch?v=R8VBRCs2jTU>

Okun's Law

- ▶ High (low) output growth is associated with a reduction (increase) in the unemployment rate.
- ▶ Changes in unemployment rate vs. output growth (US 1960-2018):



Source: FRED: Series GDPCA, UNRATE

Medium-Run Output and Inflation

Medium-Run Output and Inflation (con.)

- ▶ Over the medium run, the economy converges to the natural level of output and stable inflation.
 - ▶ Central bank has mandate to target inflation.
 - ▶ Otherwise, if inflation expectations de-anchor, PC becomes a relation between change in inflation and output gap, i.e., increasing inflation. Central bank must stop it from increasing.
- ▶ At the medium-run equilibrium, r_n is called the **natural, neutral, or Wicksellian rate of interest**.
- ▶ The nominal interest rate in the medium run is

$$i = r_n + \bar{\pi}$$

Money Neutrality

- ▶ In the medium run, real variables are independent of monetary policy: **neutrality of money**.

$$\frac{M}{P} = YL(i) = Y_n L(r_n + \bar{\pi})$$

- ▶ RHS (real money demand) is constant in steady state, so must be real money supply.
- ▶ If M grows at rate g_M , $\pi = \bar{\pi} = g_M$ and

$$i = r_n + g_M$$

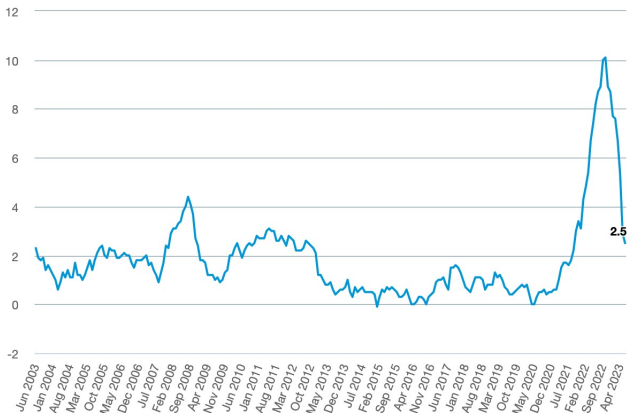
- ▶ Monetary policy determines the rate of inflation and the nominal rate of interest.

CPI in Denmark

- <https://www.dst.dk/en/Statistik/emner/oekonomi/prisindeks/forbrugerprisindeks>

Consumer price index, change compared to same month the previous year

Unit: Percentage change compared to same month the year before (per cent):



Inflation Targeting

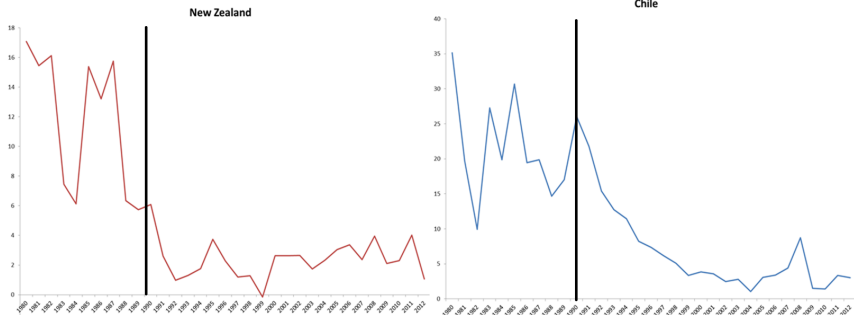
- ▶ Most central banks adjust policy interest rates expressly so as to keep inflation in check
- ▶ Many central banks now follow formal strategies of inflation targeting, under which they announce a target (or target range) for inflation rate and adjust interest rate to keep inflation on target
- ▶ Generally target “core inflation”: excludes volatile components such as energy prices
- ▶ Vs. “headline inflation”: total consumer price index

Inflation Targeting (cont.)

- ▶ Formally initiated by New Zealand's central bank in 1989
- ▶ Now, Canada, Chile, Mexico, South Africa, Sweden, Thailand, UK, euro zone, etc.
- ▶ Central banks of US and Japan strongly averse to inflation, but so far have been reluctant to announce definite inflation targets

Country	Year of adoption
Chile	1990
New Zealand	1990
Canada	1991
Finland	1993
Brazil	1999
Indonesia	2005

New Zealand and Chile



Hyperinflation

- ▶ Shrinkflation: <https://stiften.dk/debat/det-er-blevet-en-smertefuld-oplevelse-at-gaa-i-supermarkedet>
- ▶ Governments sometimes completely lose control of money
- ▶ e.g., as inflation increases, government revenue decreases, so government needs to print money to finance itself
- ▶ However, that leads to more inflation, and a vicious cycle is formed
- ▶ General reason why country ends up with hyperinflation is that the government surrenders control of money creation because it is under some imperative to raise revenue at all costs

Hyperinflation

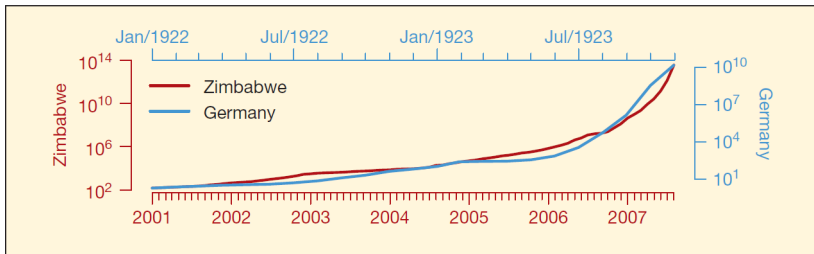


Figure 5.4 Hyperinflation in Germany and Zimbabwe

Hyperinflation

Table 5.1 Highest monthly inflation rates in history

Country	Month with highest inflation rate	Highest monthly inflation rate	Equivalent daily inflation rate	Time required for prices to double
Hungary	July 1946	$4.19 \times 10^{16}\%$	207%	15.0 hours
Zimbabwe	November 2008	$7.96 \times 10^{10}\%$	98.00%	24.7 hours
Yugoslavia	January 1994	$3.13 \times 10^8\%$	64.60%	1.4 days
Germany	October 1923	29,500%	20.90%	3.7 days
Greece	October 1944	13,800%	17.90%	4.3 days
China	May 1949	2,178%	11.00%	6.7 days

Notes. The Reserve Bank of Zimbabwe reported inflation rates for March 2007–July 2008. The authors (Hanke and Kwok, 2009, Table 2) calculated rates for August 2008–14 November 2008

Source: On the measurement of Zimbabwe's hyperinflation, *Cato Journal*, Vol. 29, No. 2, 356 (Hanke, S. H. and Kwok, A. K. F. Spring/Summer 2009). Copyright © Cato Institute. All rights reserved.

<https://www.youtube.com/watch?v=Fn76nlf92wo>

Complications and How Things Can Go Wrong

- ▶ The adjustment to the medium-run equilibrium seemed smooth and easy.
- ▶ While this process looks easy to do in the model, reality is more complicated for two reasons:
 1. It is difficult for the central bank to know where potential output is exactly and thus how far output is from potential.
 2. It takes time for the economy to respond.
- ▶ Inflation (or its change) is used as a signal of the output gap, but it is still noisy.
 - ▶ e.g., inflation is 2.3% (just above 2%), but unemployment is 3.9%
 - ▶ Is the unemployment rate too low (i.e., should increase interest rate), or natural rate has decreased?

The Deflation Spiral

- ▶ One issue that can occur is known as a **deflation spiral** or **deflation trap**.
- ▶ If the zero lower bound prevents monetary policy from increasing output back to potential, the result may be a deflation spiral.
- ▶ More deflation leads to a higher real rate, which in turn leads to lower output and more deflation.
- ▶ This scenario played out during the Great Depression.
- ▶ The recent Great Recession gave rise to similar concerns, but did not materialize.
- ▶ <https://www.youtube.com/watch?v=HvppJ-MhE6A>

The Deflation Spiral

Deflation in the Great Depression

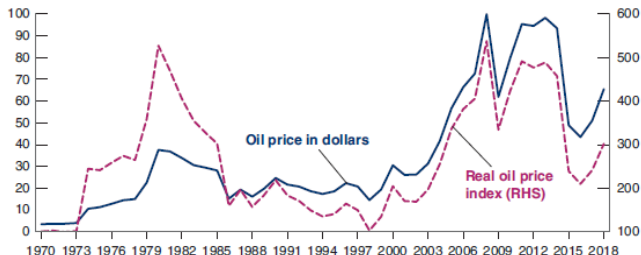
- ▶ The economy seemed to be in a deflation trap between 1929-1933.
- ▶ Monetary policy decreased the nominal interest rate from 5.3% to 2.6% in 1933, but with negative inflation rates, the real rate reached 12.3% in 1931 and 7.8% in 1933.

Year	Unemployment Rate (%)	Output Growth Rate (%)	One-Year Nominal Interest Rate (%), i	Inflation Rate (%), π	One-Year Real Interest Rate (%), r
1929	3.2	-9.8	5.3	0.0	5.3
1930	8.7	-7.6	4.4	-2.5	6.9
1931	15.9	-14.7	3.1	-9.2	12.3
1932	23.6	-1.8	4.0	-10.8	14.8
1933	24.9	9.1	2.6	-5.2	7.8

- ▶ US economy rebounded in 1934, deflation gave way to inflation; reasons remain debated.

Effects of an Increase in the Price of Oil

- Over the last 50 years, there have been two sharp increases in the real price of oil, the first in the 1970s and the second in the 2000s.



Source: FRED: OILPRICE, CPIAUSCL. Left vertical axis: nominal price of oil, dollars per barrel. Right vertical axis: real oil price index, 1970 = 100.

Effects of an Increase in the Price of Oil

- ▶ 1970s: OPEC (the Organization of Petroleum Exporting Countries) act as a monopoly and increased oil prices.
- ▶ 2000s: The fast growth of emerging economies led a rapid increase in world oil demand, and thus a steady increase in real oil prices.
- ▶ 2008: A large recession led to a sudden decrease in the demand for oil, and thus falling oil prices.
- ▶ 2014 and after: A combination of increased supply due to the increase in U.S. shale oil production and the partial breakdown of OPEC led to sudden drop in oil prices.

Effects of an Increase in the Price of Oil (con.)

Conclusions

- ▶ Shocks or changes in policy typically have different effects in the short run and in the medium run.
- ▶ Disagreements about the effects of various policies depend on how fast you think the economy adjusts to shocks.
- ▶ Movements in output around its trend are called output fluctuations (business cycles).
- ▶ Economic fluctuations are the results of shocks and their dynamic effects, called the propagation mechanism.